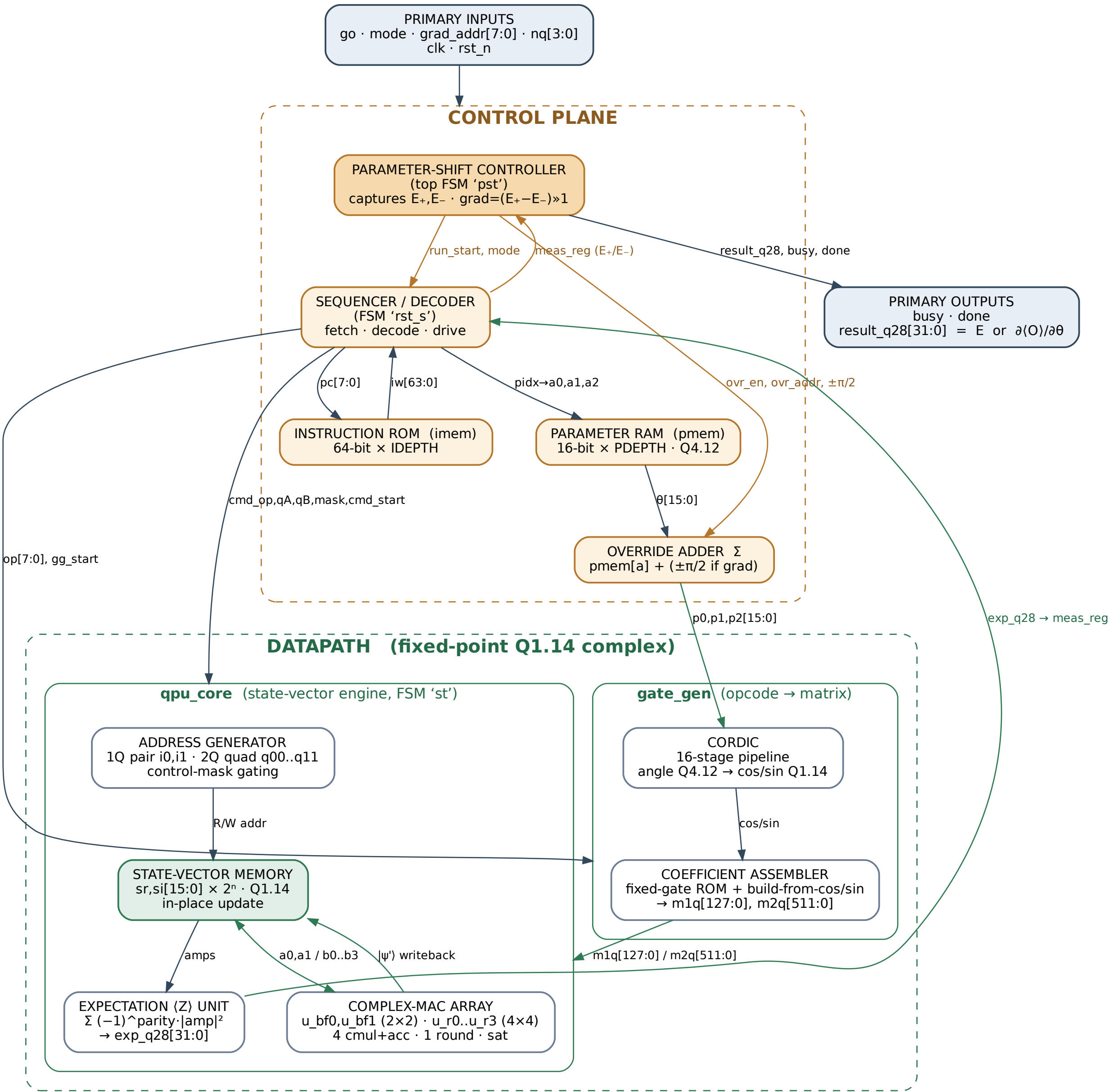


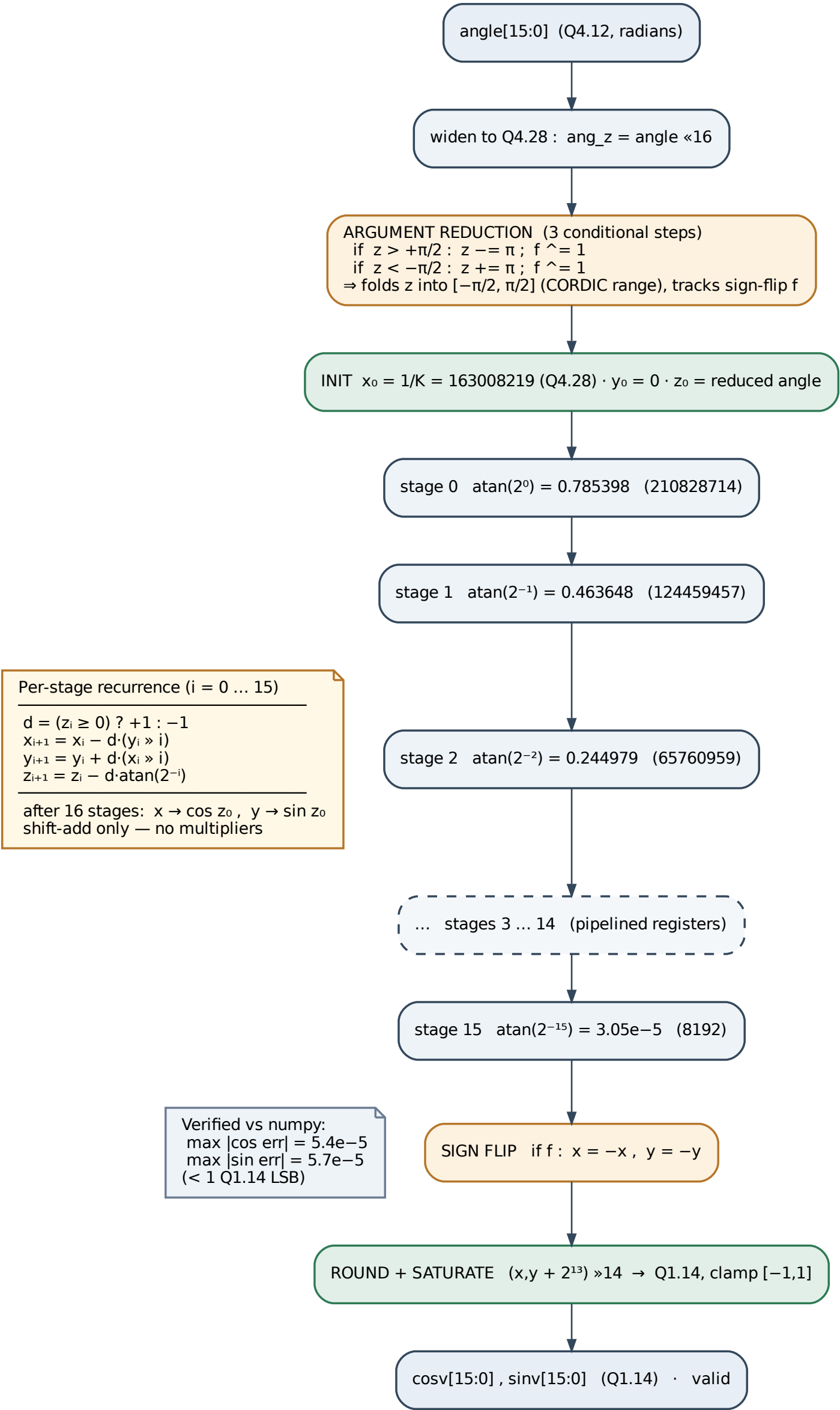
# QML Hardware Accelerator — Top-Level RTL Block Diagram

state-vector engine · module connectivity with signal names and bus widths



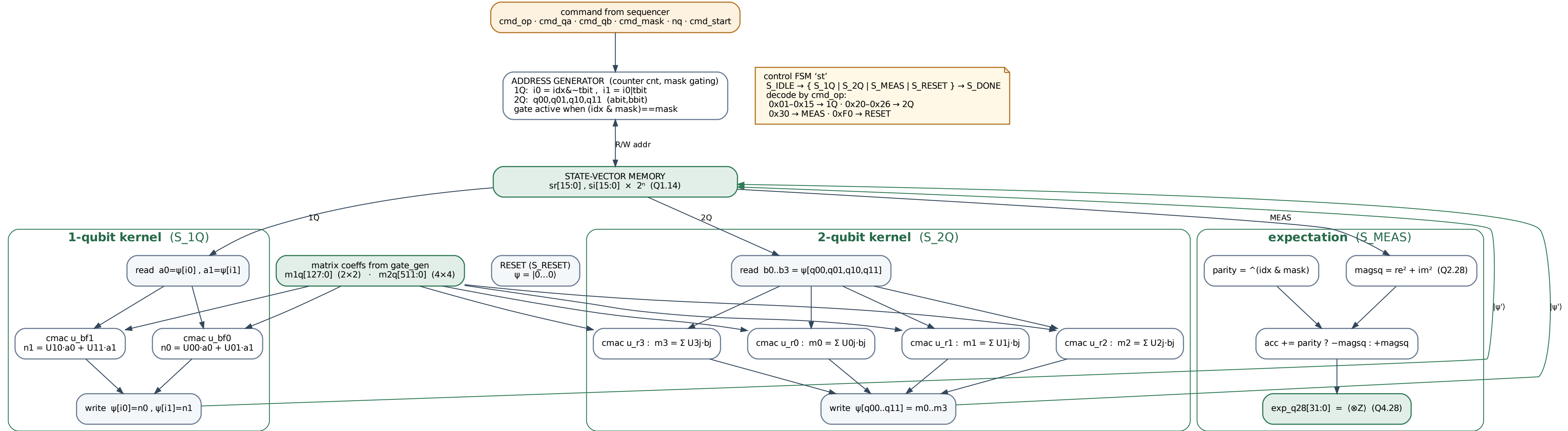
# CORDIC — Pipelined cos/sin Datapath

circular rotation mode · 16 stages · Q4.28 internal · latency = 18 cycles



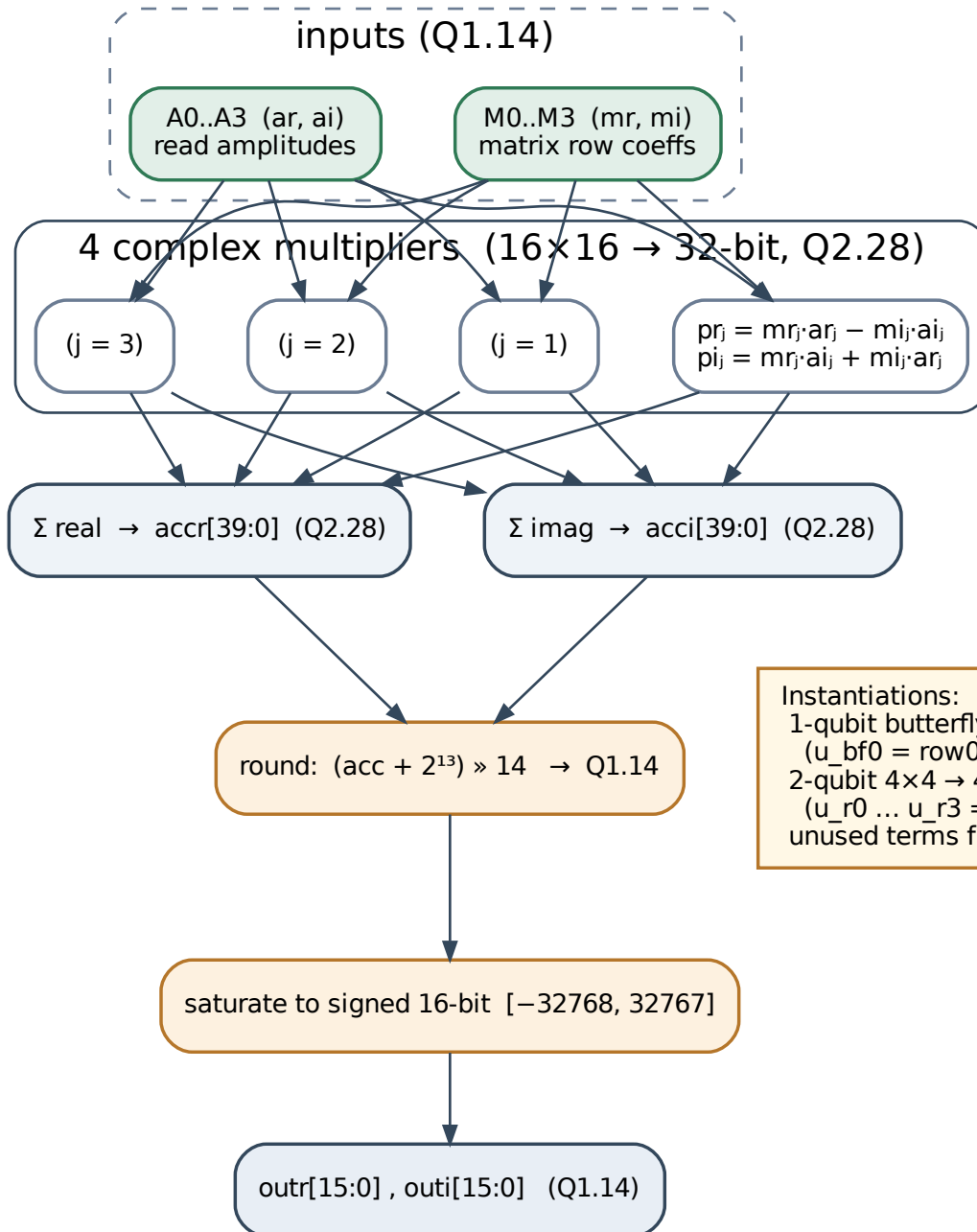
# qpu\_core — State-Vector Datapath

1-qubit butterfly · 2-qubit 4×4 kernel · ⟨Z⟩ expectation · one amplitude(-quad) per clock



# cmac — Complex Multiply-Accumulate Cell

out =  $\sum_j M_j \cdot A_j$  ( $j=0..3$ ) · full-width products · single rounding · saturating



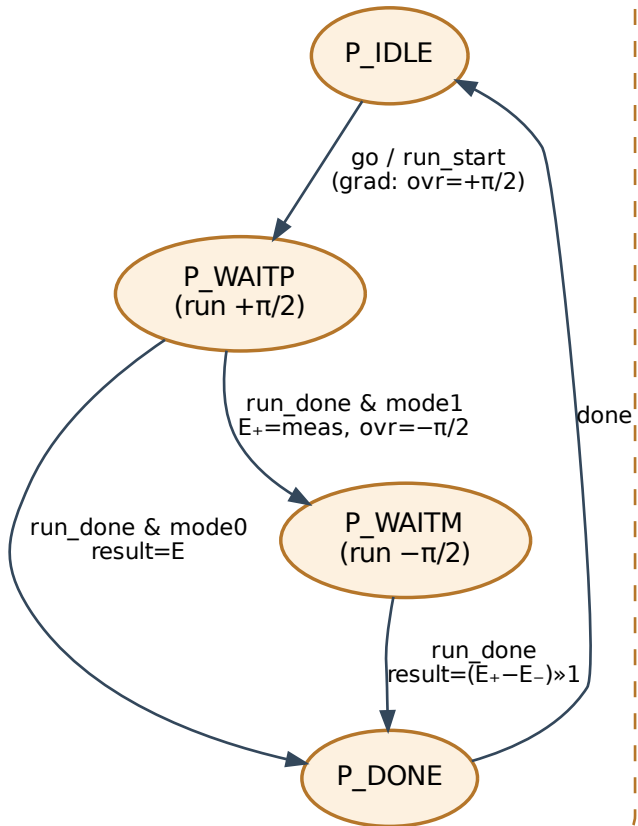
## Instantiations:

1-qubit butterfly → 2 active terms  
(u\_bf0 = row0, u\_bf1 = row1)  
2-qubit 4×4 → 4 active terms  
(u\_r0 ... u\_r3 = rows 0..3)  
unused terms fed M<sub>j</sub> = 0

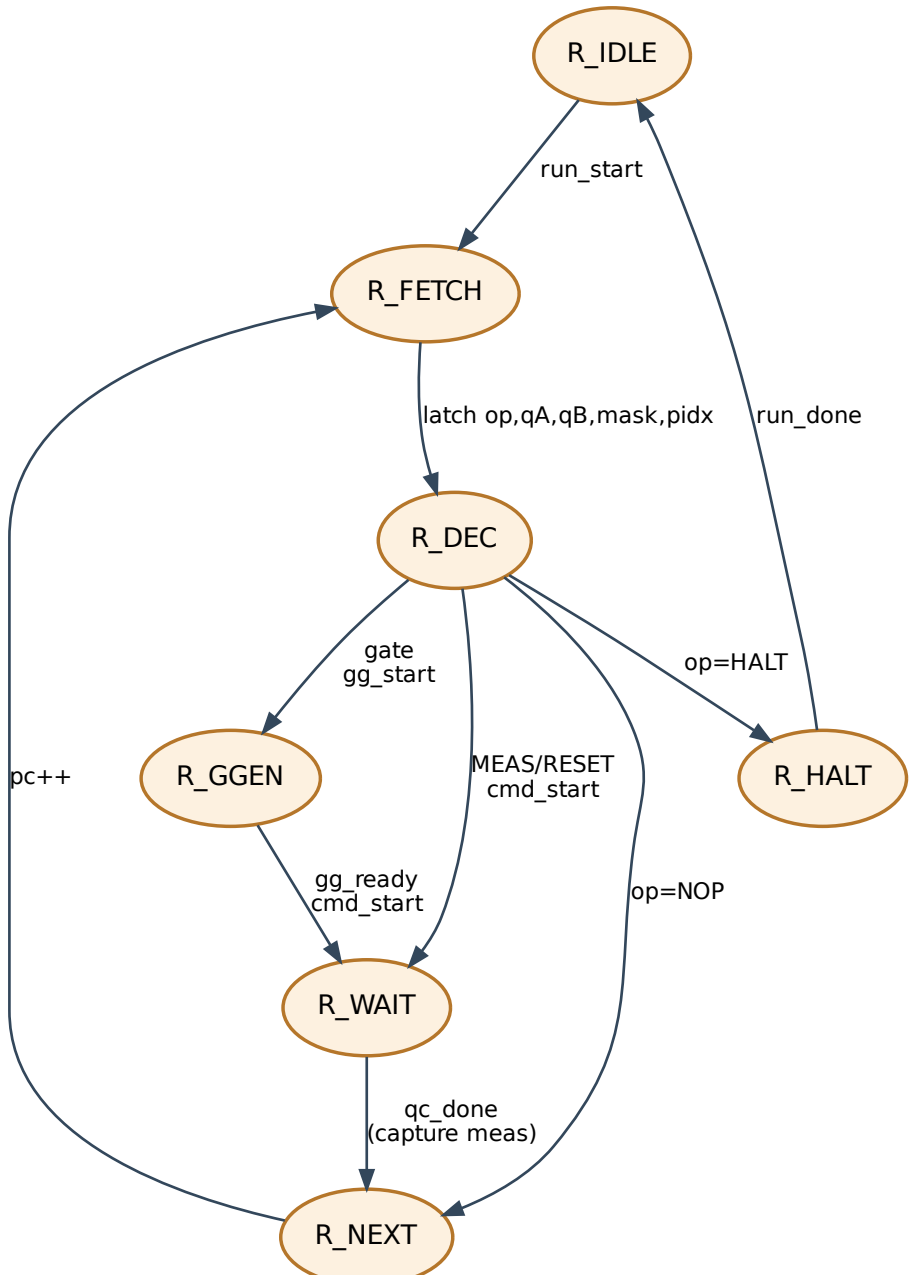
# Control Finite-State Machines

parameter-shift controller · sequencer · gate\_gen · qpu\_core

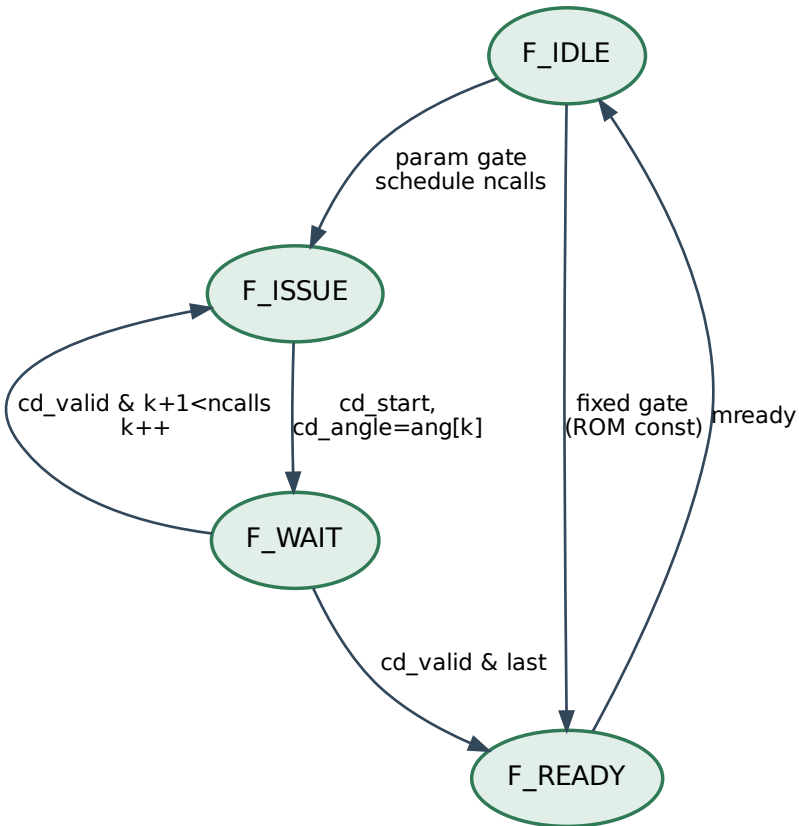
## Parameter-Shift Controller (pst)



## Sequencer / Decoder (rst\_s)



## gate\_gen (fst)



## qpu\_core (st)

